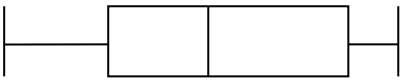
Question	Answer	Marks	Partial Marks
6(b)	[cos =] $\frac{11^2+12^2-8^2}{2\times11\times12}$	**M2**	**M1** for $8^2=11^2+12^2-2\times11\times12\cos(BAC)$
	40.415...	**A2**	**A1** for 0.761... or $\frac{201}{264}$ or $\frac{67}{88}$
6(c)	70.8 or 70.77 to 70.79...	**3**	**M1** for $\frac{1}{2}\times12\times$ *their* **(a)** $\times\sin(180-100-50)$ **M1** for $\frac{1}{2}\times12\times11\times\sin(40.42)$
6(d)	7.13 or 7.131 to 7.132...	**3**	**M2** for $\frac{\text{dist}}{11}=\sin(40.42)$ or **M1** for recognition that shortest distance is perpendicular to AC
7(a)	87	**3**	**M2** for $3c + 4c = 587 + 22$ or better or **M1** for $3c + 2(2c - 11)$ [= 587 or 5.87]
7(b)	1.1[0]	**3**	**M2** for $22w + 22 = 42w$ or better or **M1** for $\frac{22}{w} = \frac{42}{w+1}$ oe OR **B2** for number of bottles = 20 or **M1** for $Nw = 22$ and $N(w+1) = 42$
7(c)(i)	$\frac{9}{x}+\frac{5}{2x+1}=2.5$ oe	**M2**	**M1** for $\frac{9}{x}$ or $\frac{5}{2x+1}$
	$9(2x+1)+5x=2.5x(2x+1)$ oe or $\frac{9(2x+1)+5x}{x(2x+1)}$ [= 2.5 oe]	**M1**	Correctly clearing fractions, or correctly collecting into a single fraction FT *their* expression dep on two fractions both with algebraic denominators
	All brackets expanded leading to $10x^2-41x-18=0$ with no errors or omissions	**A1**	

Question	Answer	Marks	Partial Marks
7(c)(ii)	$(2x-9)(5x+2)$ or $\dfrac{-(-41)\pm\sqrt{(-41)^2-4(10)(-18)}}{2(10)}$	**M2**	**B1** for $(ax+b)(cx+d)$ with $ac = 10$ and $bd = -18$ or $ad + bc = -41$ or $\sqrt{(-41)^2-4(10)(-18)}$ or $\dfrac{-(-41)+\sqrt{q}}{2(10)}$ oe or $\dfrac{-(-41)-\sqrt{q}}{2(10)}$ oe or both or **M1** for $\left(x-\dfrac{41}{20}\right)^2-\dfrac{18}{10}-\left(\dfrac{41}{20}\right)^2=0$ or better
	10	**A2**	**A1** for $[x =]\ \dfrac{9}{2}$ oe or **M1** for $2 \times$ *their* positive root + 1
8(a)(i)	$\dfrac{60}{360}\times 600$ oe	**1**	
8(a)(ii)	45	**2**	**M1** for $\dfrac{27}{360}\times 600$ oe
8(a)(iii)	Correct straight line on the pie chart	**2**	**B1** for 75
8(b)	Correct diagram  0.6 3.4 5.2 7.5 8.7	**3**	**B1** for any three of 0.6, 3.4, 5.2, 7.5, 8.7 correctly placed **B1** for 7.5 and 8.7 seen
8(c)(i)	5	**1**	
8(c)(ii)	2	**1**	
8(c)(iii)	3	**1**	
8(d)	39.2	**4**	**M1** for mid-values soi **M1** for Σfx with x in correct interval including boundaries **M1 dep** for $\dfrac{\Sigma fx}{50}$ dep on second M1
9(a)	(0, 0), (1, 0), (2, 0)	**2**	**B1** for any two correct If 0 scored, SC1 for all three x values clearly identified